

Weak Near-Threshold Interactions in $D^* \bar{D}$ Scattering from Lattice QCD

ASCR/NP LQCD SciDAC-5

Scientific Achievement

- Spectrum of charmonium quark spectrum of Quantum Chromodynamics - subject of intense experimental scrutiny
- First-principles lattice QCD study of coupled $\eta_c^* \eta_c$ and $D^* \bar{D}$ scattering
- No evidence found for exotic charmonium resonance states near $D^* \bar{D}$ thresholds in contradiction to experimental determinations

Significance and Impact

- Clean first-principles QCD baseline suggesting several experimental near-threshold “exotic” candidates may arise from production effects rather than genuine strong $D^* \bar{D}$ dynamics
- Challenges molecular/tetraquark interpretations
- Supports experimental programs (LHCb, Belle II, BESIII)
- Advances lattice QCD as precision tool
- Enables HPC-driven nuclear theory insights

Technical Approach

- Calculations used HPC systems at OLCF and JLab
- Utilizes finite-volume coupled-channel scattering amplitude formalism
- Correlation function codes developed under SciDAC

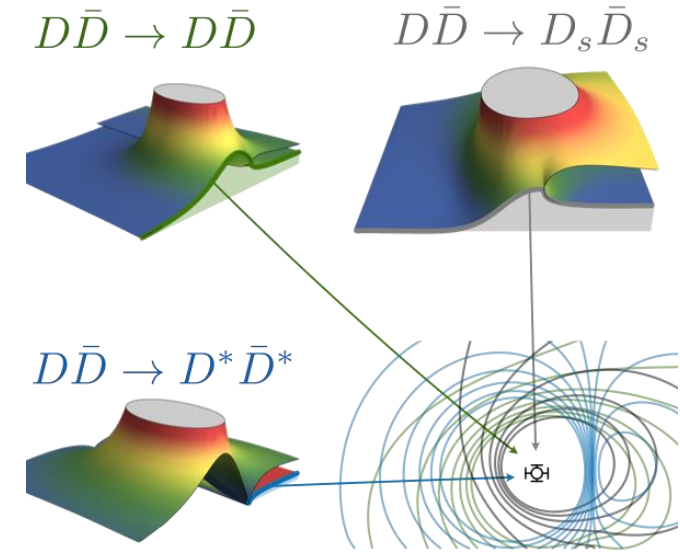


Figure representing Lattice QCD calculations of the spectrum of charm quarks – the subject of intense experimental scrutiny

PI(s)/Facility Lead(s): [D.J. Wilson, J.J. Dudek, C.T. Thomas, R.G. Edwards]

Collaborating Institutions: [William and Mary, Cambridge U.]

ASCR Program: [ERCAP, INCITE, SciDAC]

ASCR PM: [Xiaofeng Guo]

Publication(s) for this work: D.J. Wilson, et al., “Weak Near-Threshold Interactions in $D^* \bar{D}$ Scattering from Lattice QCD,” arXiv:2602.09862; Phys.Rev.D 114 (2026) 5, 054028